Topics of today's class

Last class:

- Chapter 4
  - Consumer and Firm Behavior
  - The Work-Leisure Decision and Profit Maximization

Today's class:

- Chapter 5
  - Introduce the government.
  - Construct closed-economy one-period macroeconomic model.
  - Economic efficiency and Pareto optimality

Chapter 5

A Closed Economy
One-Period Macroeconomic Model

Chapter 5 Topics

- Introduce the government.
- Construct closed-economy one-period macroeconomic model.
- Economic efficiency and Pareto optimality.
- Experiments:
  - Increases in government spending and total factor productivity.
  - Consider a distortionary tax on wage income and study the Laffer curve.

Closed-Economy One-Period Macro Model

- Representative Consumer
  - decides how much to work and how much to consume: (c, N_s) taking as given the wage rate w and the interest rate r.

- Representative Firm
  - decides how much to produce Y and how much labor to hire N_d taking as given the wage rate w and the interest rate r.

- Government
  - raises taxes T and spends G.

- Static world: K *is* constant.

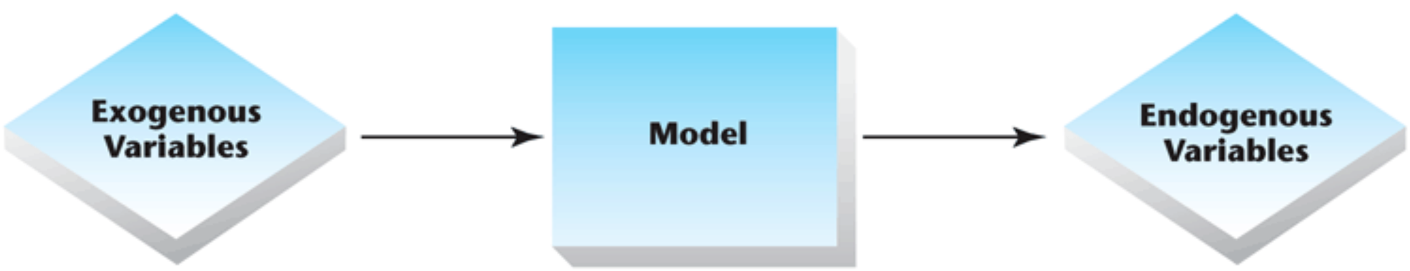
Why?

We will now put these together. Why?

1. Consistency: we are sure that everyone is doing things that are compatible.
2. To derive positive predictions from the model.

Exogenous Variables and Endogenous Variables

Exogenous Variables: K, G, z.
Endogenous Variables: w, r, N, c, T, Y, π, l.



Competitive Equilibrium

A Competitive Equilibrium is

1. an allocation $\{Y, N, K, c\}$;
2. a price system $\{w, r\}$;
3. a government policy $\{T, G\}$.

such that

- Representative consumer optimizes given market prices.
- Representative firm optimizes given market prices.
- The government budget constraint is satisfied, or $G = T$.
- Markets clear: goods market, labor market and capital market.

Income-Expenditure Identity

In a competitive equilibrium, the income-expenditure identity is satisfied, so

$$Y = C + G$$

why?

Income-Expenditure Identity

Consumer's Budget constraint:

$$C = wN_s + rK + \pi - T$$

Profits:

$$\pi = Y - wN_d - rK$$

Government Budget Constraint:

$$T = G$$

Income-Expenditure Identity

Consumer's Budget constraint:

$$C = wN_s + Y - wN_d - G$$

and Equilibrium on the Labor Market

$$N_s = N_d$$

done!

Solving for an equilibrium

- Graphical Approach
- Key here: to find $\boldsymbol{w}$ *(wage)* that clears labor market.

The Production Function

Production Function

$$Y = zF(K,N)$$

Time Constraint

$$N = h - l$$

So

$$Y = zF(K, h - l)$$

Production Possibilities Frontier

In Equilibrium

$$C = Y - G$$

combined with

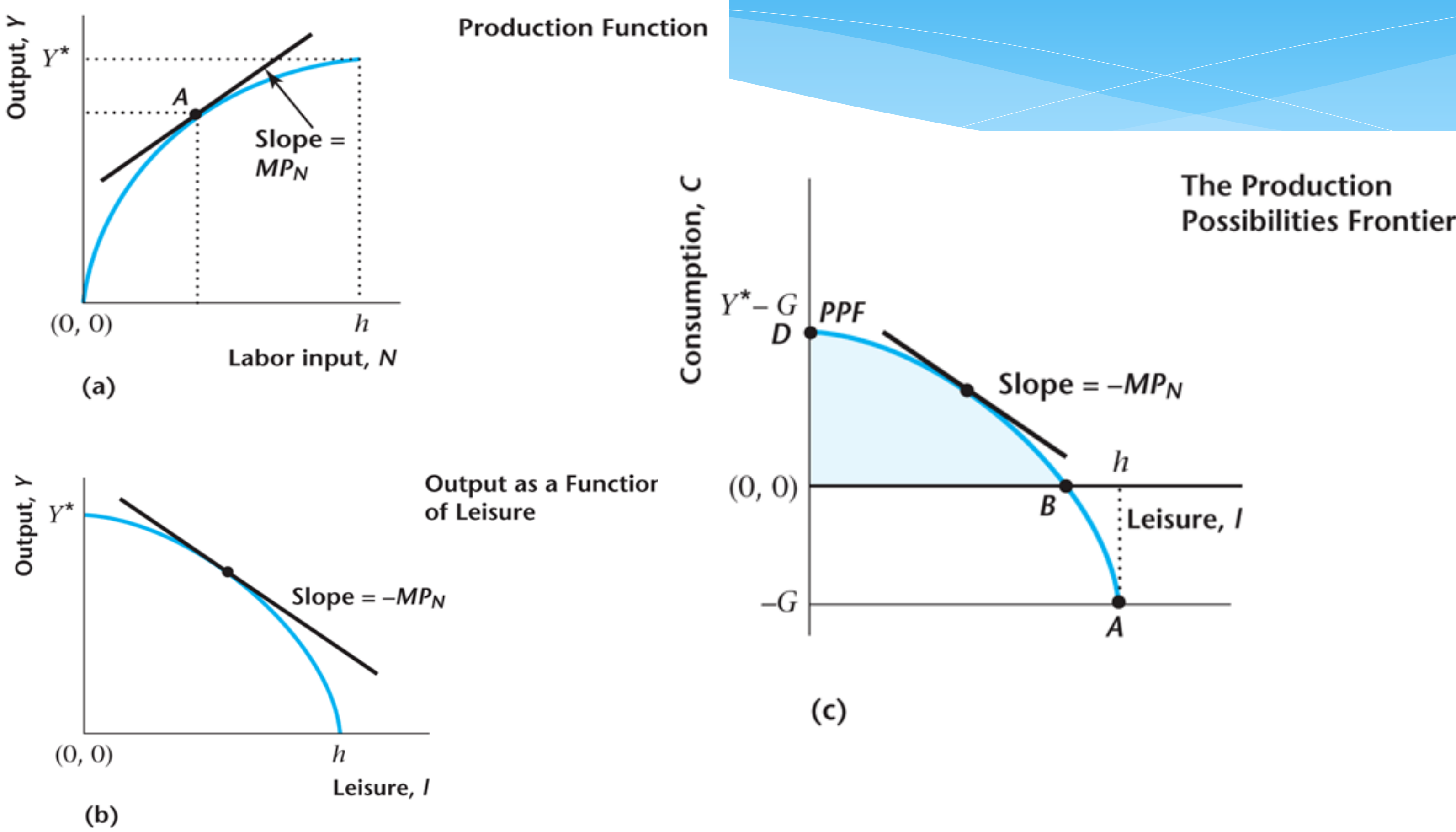
$$Y = zF\,(K,\, h - l)$$

finally:

$$C = zF\,(K,\, h - l) - G$$

Definition: The ***Production Possibilities Frontier*** describes what the technological possibilities are, for the economy as the whole, in terms of the production of consumption goods and leisure.

Figure 5.2
The Production Function and the Production Possibilities Frontier



Marginal Rate of Transformation

- The marginal rate of transformation (MRT) is the rate at which one good can be converted technologically into another.
- In our case, the MRT is the rate at which leisure can be converted in the economy into consumption goods through work.
- $MRT_{l,c}$ = MPN = -(the slope of the PPF).

Key Properties of a Competitive Equilibrium

From consumer utility maximization:

$$MRS_{l,C} = w$$

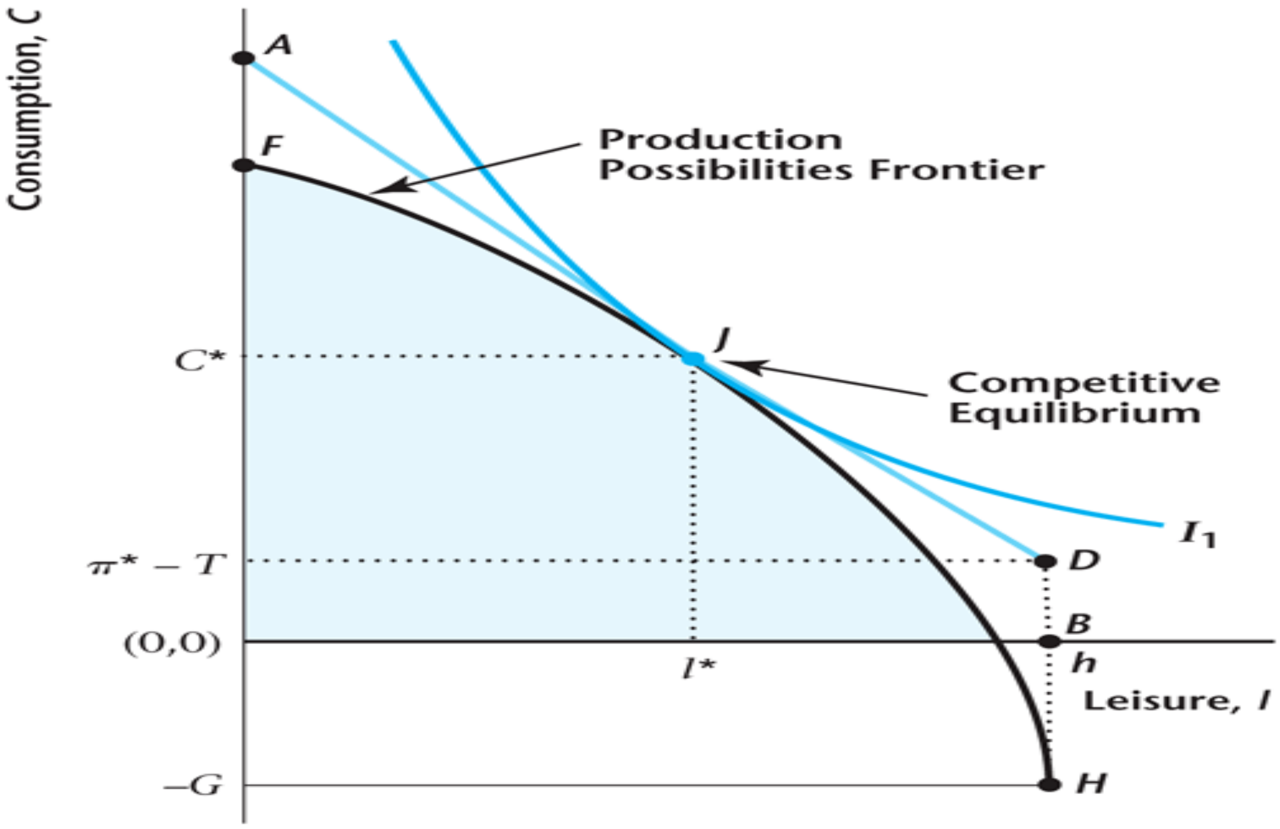
From firm profit maximization:

$$MPN = w$$

Imposing labor market clearing:

$$MRS_{l,C} = MPN$$

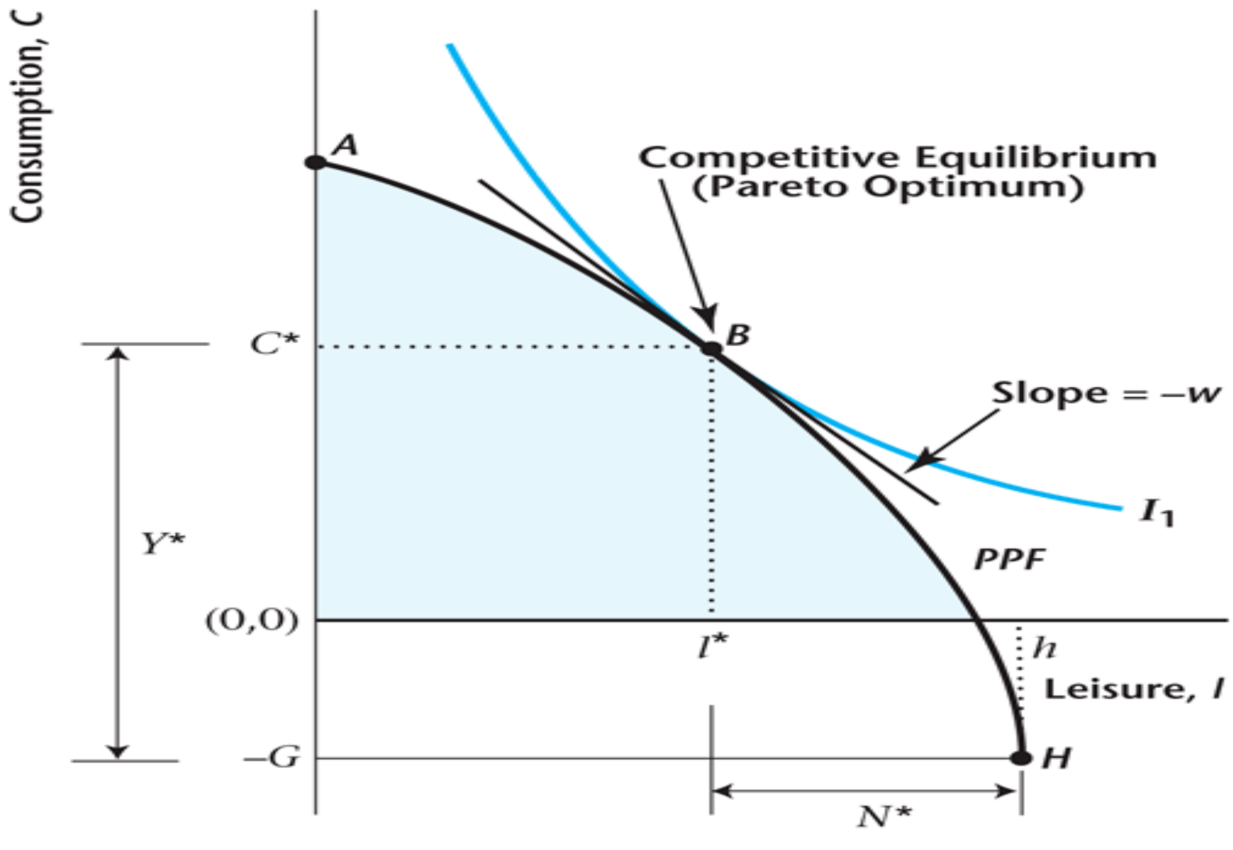
Figure 5.3
Competitive Equilibrium



Pareto Optimality

- A competitive equilibrium is ***Pareto Optimal*** if there is no way to rearrange production or to reallocate goods so that someone is made better off without making someone else worse off.

Figure 5.4
Pareto Optimality



Key Properties of a Pareto Optimum

- In this model, the competitive equilibrium and the Pareto optimum are identical.

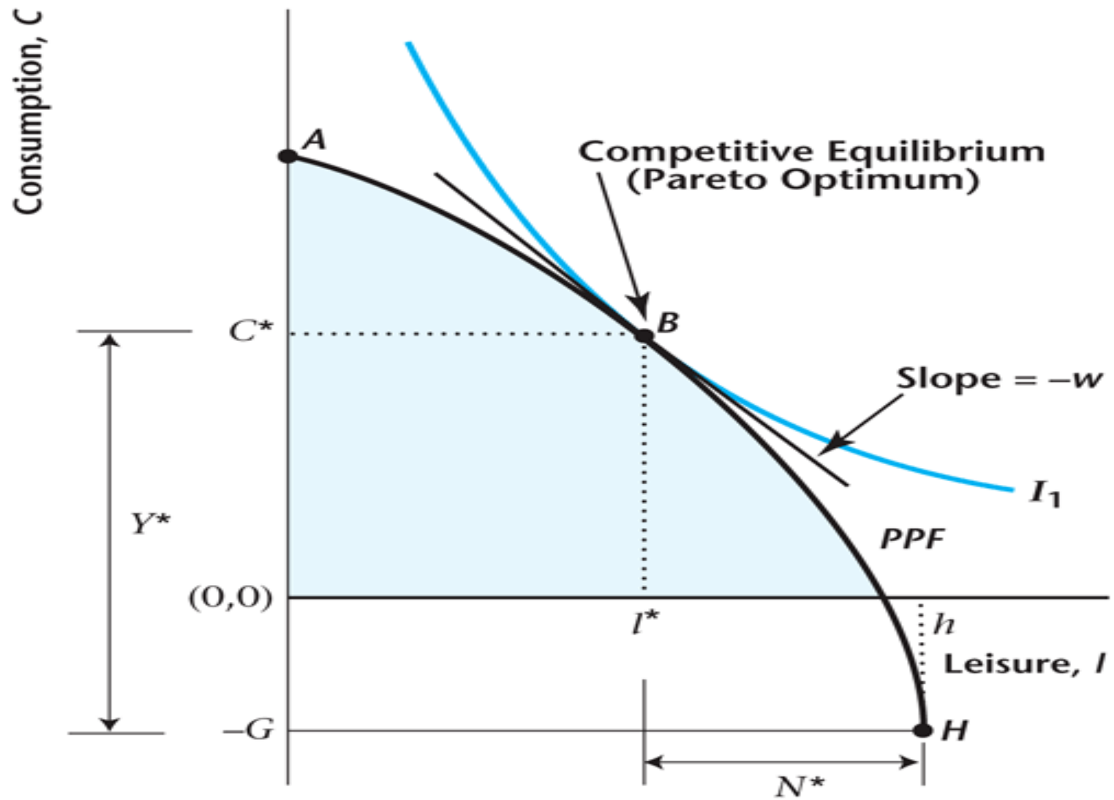
- We know this as, at the Pareto optimum,

$$MRS_{l,C} = MRT_{l,C} = MP_N$$

First and Second Welfare Theorems

- These theorems apply to any macroeconomic model.
- First Welfare Theorem: Under certain conditions, a competitive equilibrium is Pareto optimal.
- Second Welfare Theorem: Under certain conditions, a Pareto optimum is a competitive equilibrium.

Figure 5.5
Using the Second Welfare Theorem to Determine a Competitive Equilibrium



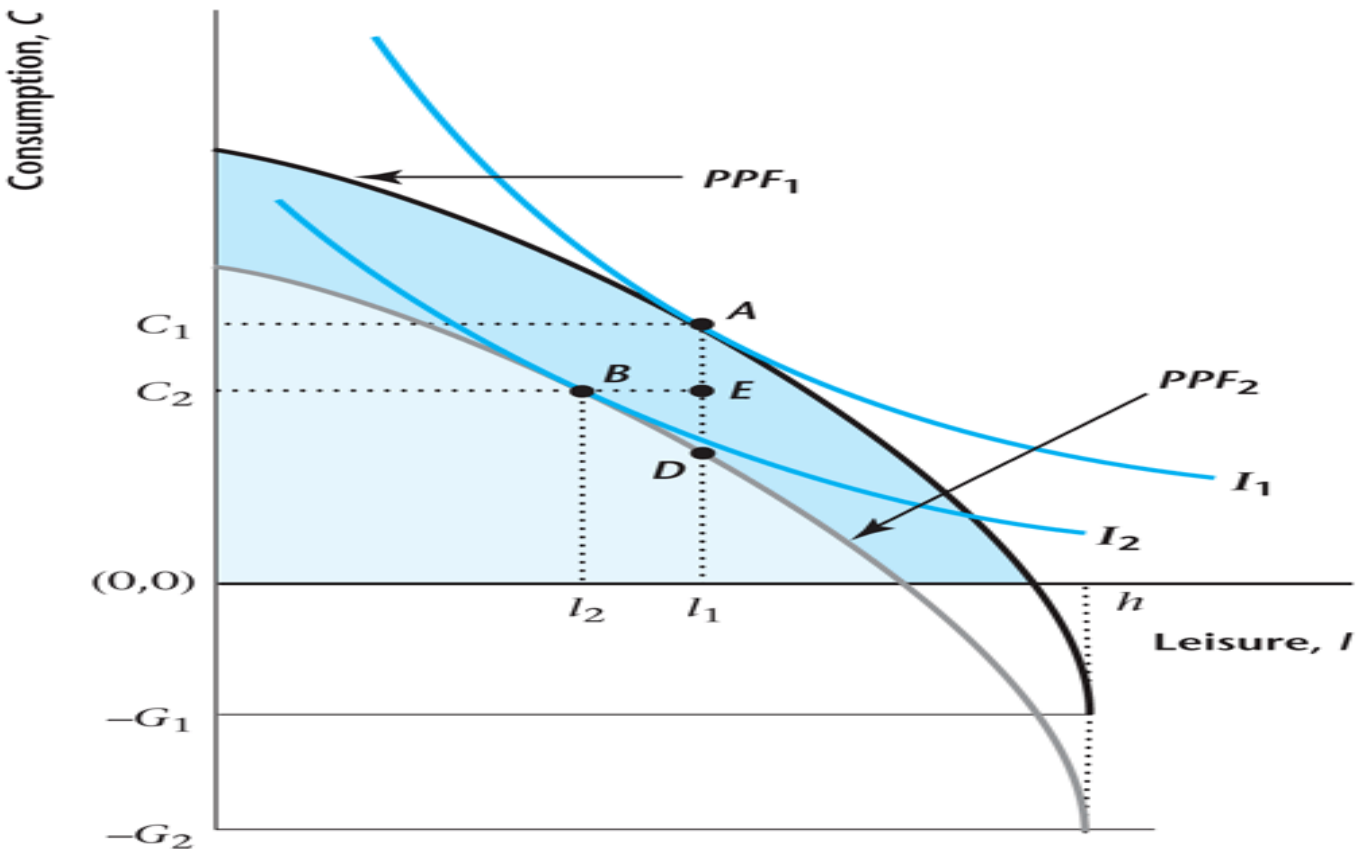
Thought Experiments

- Increase in Governments Expenditures
  - Government Expenditures during World War II
- Increases in total factor productivity
- Impact of a tax proportional to wages

WWII and increase in G

- During WWII government spending to finance the war effort increased to levels unseen previously in the US.
- What are the predictions of the model for this increase in spending?
- The assumption that government spending is a pure loss of output arguably makes sense here.
  - Military expenditure protect us.
  - But means less consumption goods and services.

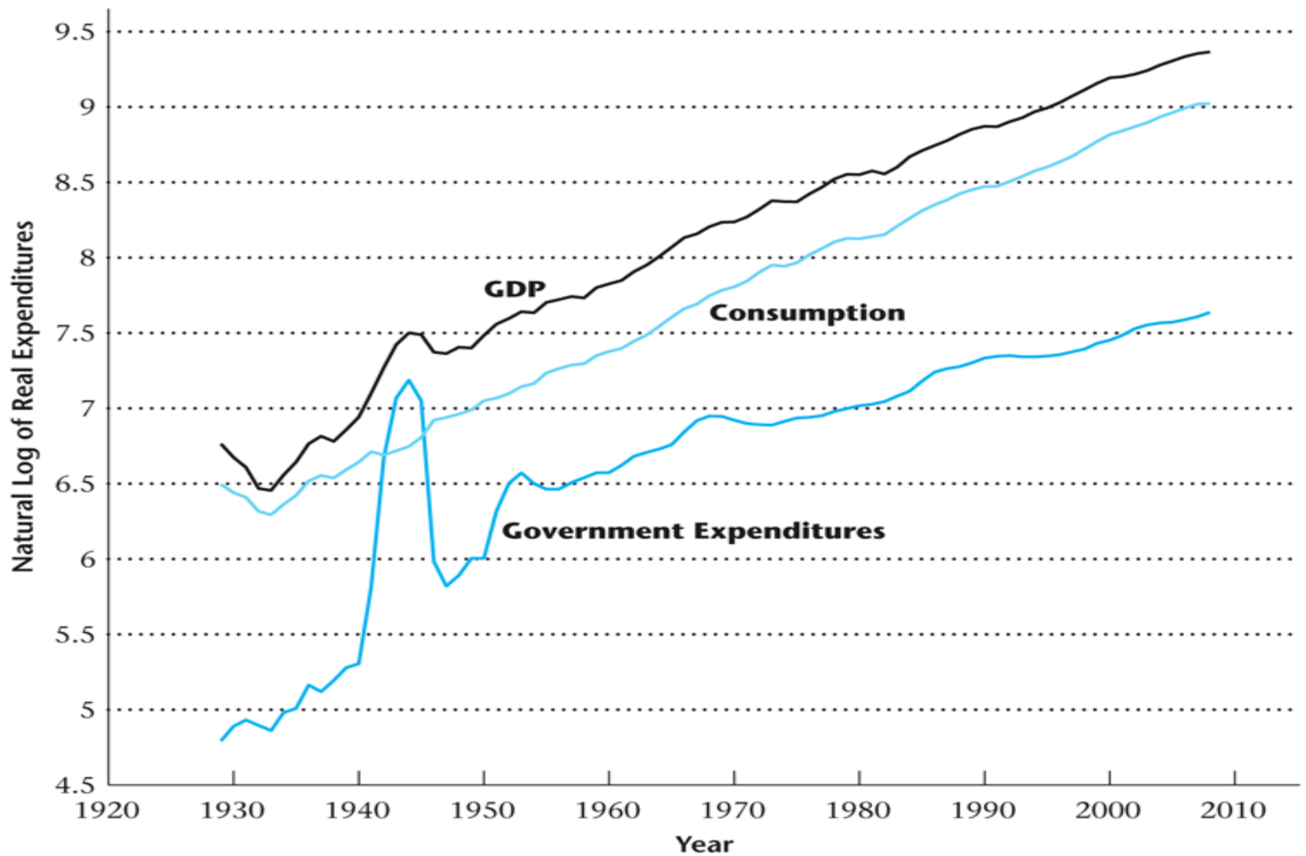
Figure 5.6
Equilibrium Effects of an Increase in Government Spending



World War II Increase in G

- Essentially: Negative Income Effects
  - increase in (Y, N),
  - decrease in (c, w).
- Private consumption spending is "crowded out" by increased government spending.
- Loss of welfare as both c, l fall.
  - Crucial Assumption: G is not in the Utility Function
- These predictions match U.S. experience of WWII.

Figure 5.7. GDP, Consumption, and Government Expenditures



WWII

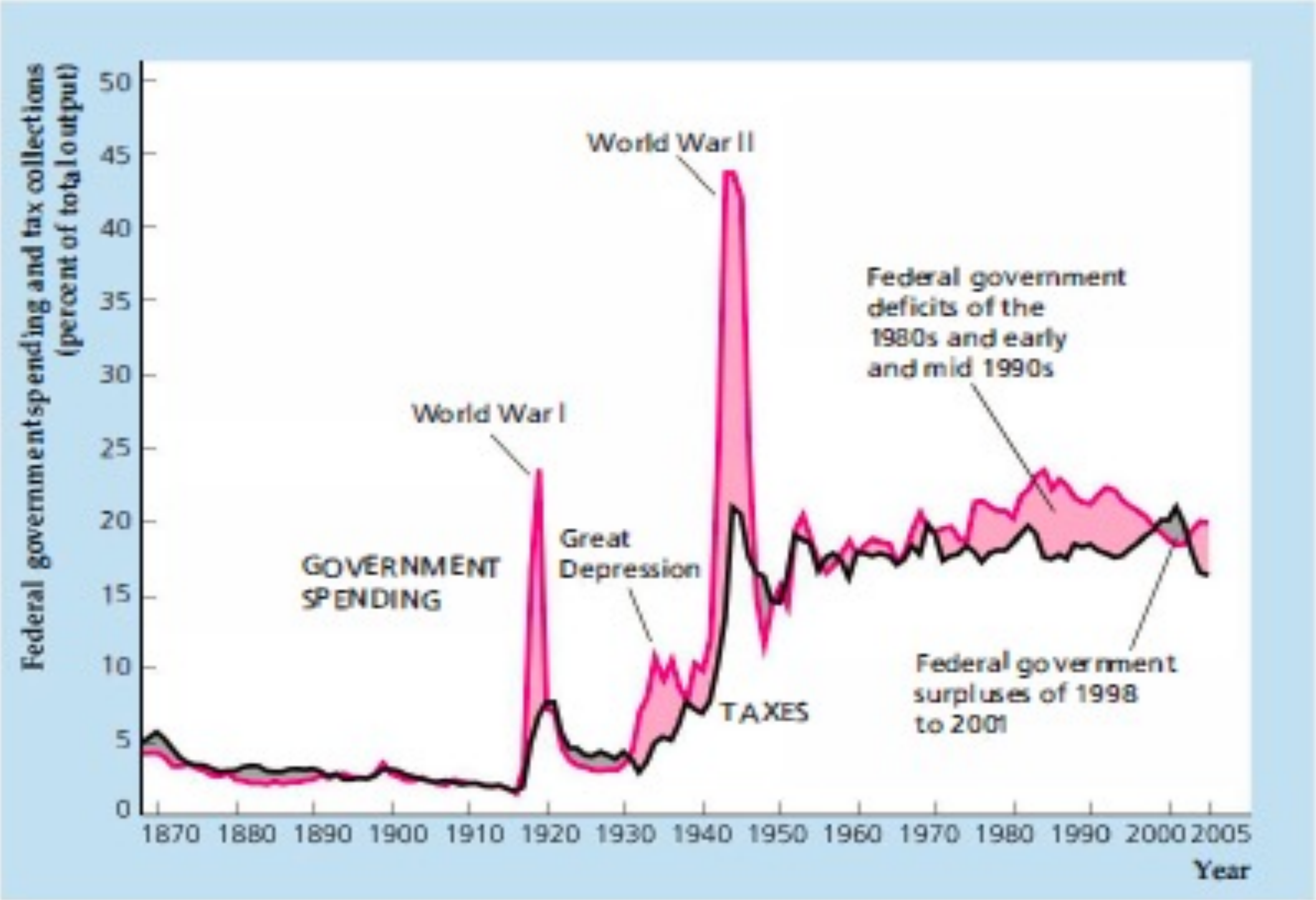
- Large Increase in Government Expenditures.
- Y increases.
- C decreases by a small amount.

What does this analysis miss?

- Government debt. A large fraction of the wartime spending was financed by government debt.
  - Deficit/GDP ratio hit 24% by 1944.
  - So Taxes increased less which is maybe why consumption only decreased by a small amount

- Increased productivity. Wartime mobilization of production increased labor productivity dramatically.
  - Increase in Output is larger than our model suggests

Deficit



Federal government spending and tax collections (percent of total output)
50
45
40
35
30
25
20
15
10
5
0
World War II
Federal government deficits of the 1980s and early and mid 1990s
World War I
Great Depression
GOVERNMENT SPENDING
TAXES
Federal government surpluses of 1998 to 2001
1870 1880 1890 1900 1910 1920 1930 1940 1950 1960 1970 1980 1990 2000 2005
Year

Labor Productivity